Balance Table

Covariate balance and wave composition (companion to Causal_Forest.Rmd)

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Contents

1 Setup and data preparation **1**

2 Covariate balance table **2**

3 Wave checks **3**

3.1 Wave composition 3

3.2 Balance within wave 4

4 Joint balance test **5**

Balance checks for the causal forest sample. They do not depend on any forest, so they live here rather than in `Causal_Forest.Rmd`. Both files read the analysis sample saved by `cleaned_data.Rmd` (`cleaned data.Rda`), so knit that one first.

1 Setup and data preparation

```
required_packages <- c("estimatr", "dplyr", "knitr")
missing_packages  <- setdiff(required_packages, rownames(installed.packages()))
if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}
library(estimatr)
library(dplyr)
library(knitr)
```

```
# cleaned_data.Rmd builds and saves the analysis sample (df_int1)
rda_path <- "cleaned data.Rda"
if (!file.exists(rda_path)) {
  stop("Cleaned data not found. Knit cleaned_data.Rmd first to create: ",
       normalizePath(rda_path, mustWork = FALSE))
}
loaded <- load(rda_path)
stopifnot("df_int1" %in% loaded)
```

```r
# Pre-treatment resume characteristics
all_pretreat_cov <- c(
  "language_skills",
  "female",
  "ba_quality",
  "extracurricular_skills",
  "ma",
  "same_exp",
  "exp_highquality"
)

# Missing values are NOT recoded in cleaned_data.Rmd: same_exp is NA for every resume in
# wave 2. The balance statistics below therefore use the observed resumes only, so the
# same_exp row covers waves 0-1.
# The wave is saved as a factor; use a numeric code (0, 1, 2).
df_int1 <- df_int1 %>%
  mutate(fall_data = as.numeric(as.character(fall_data)))

waves <- sort(unique(df_int1$fall_data))
```

2 Covariate balance table

Standardised Mean Difference (SMD) for covariate X = (mean in D=1 minus mean in D=0) / sqrt((var in D=1 + var in D=0) / 2). Rule of thumb: |SMD| < 0.1 counts as acceptable covariate balance and |SMD| >= 0.1 is flagged for checks. The p-value is from a firm-clustered regression of each covariate on D. Wave is a three-level code, so it enters as one dummy per wave rather than as a numeric code. same_exp is only observed in waves 0-1 (the `missing` column counts the resumes without it), so its row is calculated on those waves only. N counts the number of resumes with X = 1 within each treatment group.

```r
smd <- function(x, d) {
  x1 <- x[d == 1]
  x0 <- x[d == 0]
  (mean(x1, na.rm = TRUE) - mean(x0, na.rm = TRUE)) /
    sqrt((var(x1, na.rm = TRUE) + var(x0, na.rm = TRUE)) / 2)
}

# One dummy per wave
df_bal <- df_int1
for (k in waves) df_bal[[paste0("wave_", k)]] <- as.numeric(df_bal$fall_data == k)

bal_vars <- c(all_pretreat_cov, paste0("wave_", waves))

balance_row <- function(v, dat) {
  x <- as.numeric(dat[[v]])
  p <- tryCatch({
    # lm_robust drops resumes where the variable is missing
    fit <- lm_robust(as.formula(paste(v, "~ D")), data = dat,
                     clusters = firmid, se_type = "CR2")
    unname(fit$p.value["D"])
  }, error = function(e) NA_real_)
  # Counts only make sense for 0/1 variables
  is_bin <- all(x[!is.na(x)] %in% c(0, 1))
```

```
  data.frame(variable = v,
             n_D0     = if (is_bin) sum(x[dat$D == 0] == 1, na.rm = TRUE) else NA_integer_,
             n_D1     = if (is_bin) sum(x[dat$D == 1] == 1, na.rm = TRUE) else NA_integer_,
             mean_D0  = mean(x[dat$D == 0], na.rm = TRUE),
             mean_D1  = mean(x[dat$D == 1], na.rm = TRUE),
             SMD      = smd(x, dat$D),
             p_clustered = p,
             n_missing = sum(is.na(x)))
}

bal_tab <- bind_rows(lapply(bal_vars, balance_row, dat = df_bal)) %>%
  mutate(flag = ifelse(abs(SMD) >= 0.1, "check", ""))

kable(bal_tab, digits = 3,
      col.names = c("variable", "N D=0", "N D=1", "mean D=0", "mean D=1",
                    "SMD", "p (clustered)", "missing", "flag"),
      caption = paste("Balance of resume characteristics by name type",
                      "(D = 1 foreign name)"))
```

Table 1: Balance of resume characteristics by name type (D = 1 foreign name)

variable	N D=0	N D=1	mean D=0	mean D=1	SMD	p (clustered)	missing	flag
language_skills	748	676	0.247	0.257	0.022	0.384	0	
female	1544	1332	0.510	0.506	-0.008	0.767	0	
ba_quality	1784	1673	0.590	0.636	0.095	0.000	0	
extracurricular_skills	1817	1600	0.600	0.608	0.016	0.593	0	
ma	620	522	0.205	0.198	-0.016	0.547	0	
same_exp	859	639	0.433	0.452	0.037	0.250	2260	
exp_highquality	1001	836	0.331	0.318	-0.028	0.328	0	
wave_0	953	949	0.315	0.361	0.097	0.000	0	
wave_1	1029	466	0.340	0.177	-0.379	0.000	0	check
wave_2	1044	1216	0.345	0.462	0.241	0.000	0	check

3 Wave checks

The balance table shows treated share differs a lot by wave. These two checks ask whether that is a composition issue (which name types were sampled in which wave) or a randomisation problem within wave. (Whether the name penalty itself differs by wave is checked in `Causal_Forest.Rmd`.)

3.1 Wave composition

```
# Which name types appear in which wave?
print(table(wave = df_int1$fall_data,
            ethnicity = as.character(df_int1$name_ethnicity)))
```

```
##     ethnicity
## wave Canada Chinese Indian Pakistani
```

```
##     0    953     481    468         0
##     1   1029       0      0       466
##     2   1044     517    699         0
```

```r
# Treatment share and callback rate by wave
wave_tab <- df_int1 %>%
  group_by(wave = fall_data) %>%
  summarise(n = n(),
            treated_share    = mean(D),
            callback_rate_D0 = mean(as.numeric(callback)[D == 0]),
            callback_rate_D1 = mean(as.numeric(callback)[D == 1]),
            .groups = "drop")

kable(wave_tab, digits = 3, caption = "Treatment share and callback rate by wave")
```

Table 2: Treatment share and callback rate by wave

wave	n	treated_share	callback_rate_D0	callback_rate_D1
0	1902	0.499	0.158	0.115
1	1495	0.312	0.127	0.079
2	2260	0.538	0.141	0.098

3.2 Balance within wave

```r
# Difference between name types after absorbing wave fixed effects.
# Resumes where the variable is missing are dropped (same_exp: waves 0-1 only).
within_row <- function(v) {
  dat   <- df_int1
  dat$x <- as.numeric(dat[[v]])
  fit   <- lm_robust(x ~ D + factor(fall_data), data = dat,
                     clusters = firmid, se_type = "CR2")
  data.frame(variable         = v,
             diff_D1_minus_D0 = unname(coef(fit)["D"]),
             se               = unname(fit$std.error["D"]),
             p_clustered      = unname(fit$p.value["D"]))
}

within_tab <- bind_rows(lapply(all_pretreat_cov, within_row))

kable(within_tab, digits = 3,
      caption = paste("Balance within wave (name-type difference,",
                      "wave fixed effects, firm-clustered)"))
```

Table 3: Balance within wave (name-type difference, wave fixed effects, firm-clustered)

variable	diff_D1_minus_D0	se	p_clustered
language_skills	-0.003	0.011	0.811
female	-0.001	0.014	0.963
ba_quality	0.015	0.012	0.227
extracurricular_skills	0.007	0.015	0.609

variable	diff_D1_minus_D0	se	p_clustered
ma	-0.008	0.011	0.468
same_exp	0.000	0.016	0.977
exp_highquality	0.010	0.014	0.473

4 Joint balance test

A single test of whether the resume characteristics jointly predict the name type. D is regressed on all covariates plus wave fixed effects with firm-clustered (CR2) standard errors, and the test is that every covariate coefficient is zero (a Wald test, reported as F = chi-square / k, with the number of firms minus one as the denominator degrees of freedom). A large p-value means there is no evidence that the characteristics were assigned in relation to the name type. same_exp is only observed in waves 0-1, so the test is run twice: on all waves without same_exp, and on waves 0-1 with it.

```
joint_balance <- function(vars, label) {
  dat <- df_int1[complete.cases(df_int1[, vars]), ]
  fit <- lm_robust(as.formula(paste("D ~", paste(vars, collapse = " + "),
                                    "+ factor(fall_data)")),
                   data = dat, clusters = firmid, se_type = "CR2")
  b     <- coef(fit)[vars]
  V     <- vcov(fit)[vars, vars]
  wald  <- as.numeric(t(b) %*% solve(V) %*% b)
  k     <- length(vars)
  firms <- length(unique(dat$firmid))
  data.frame(sample  = label,
             resumes = nrow(dat),
             firms   = firms,
             k       = k,
             F_stat  = wald / k,
             p       = pf(wald / k, k, firms - 1, lower.tail = FALSE))
}

joint_tab <- bind_rows(
  joint_balance(setdiff(all_pretreat_cov, "same_exp"),
                "All waves, without same_exp"),
  joint_balance(all_pretreat_cov,
                "Waves 0-1, with same_exp")
)

kable(joint_tab, digits = 3,
      col.names = c("sample", "resumes", "firms", "covariates", "F", "p"),
      caption = "Joint balance test (D on all covariates + wave FE, firm-clustered)")
```

Table 4: Joint balance test (D on all covariates + wave FE, firm-clustered)

sample	resumes	firms	covariates	F	p
All waves, without same_exp	5657	3269	6	0.456	0.841
Waves 0-1, with same_exp	3397	1927	7	0.353	0.929